

Childcare Affordability and Data Visualization

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Summary of Analysis

This project shows the issues of being able to afford childcare across the United States using the *National Database of Childcare Prices (NDCP)*. The primary goal was to identify trends in childcare costs relative to median household income and highlight the struggles within different age groups and geographic locations. Using different types of media, including PowerPoint presentations, a community poster, and a public blog post.

Using Python, the dataset was cleaned, standardized, and visualized through various charts, including bar graphs, boxplots, heatmaps, and choropleth maps. These visualizations serve as the foundation for communicating complex data to diverse audiences, demonstrating both the economic and regional challenges families face in accessing affordable childcare.

Findings

The analysis revealed several important insights about childcare affordability. First, infant care consistently represents the highest financial burden, with families spending up to 12% of median household income, and in some counties exceeding 20%. Toddler and preschool care costs are lower, but still substantial. Geographic disparities were also apparent, with states such as Massachusetts, California, and Washington being the least affordable, while Mississippi, South Dakota, and Arkansas were among the most affordable. Outlier counties with unusually high or low costs were often associated with workforce shortages, higher living expenses, or limited access to providers, particularly in rural regions. These patterns underscore the intersection of age-specific costs, income inequality, and regional disparities in determining childcare affordability.

Assumptions

Several assumptions were made during the analysis to ensure. During the analysis, missing or incomplete values were treated as unreported instead of zero. Data was grouped by state to show overall trends, and extreme outliers were removed to avoid misleading charts. All these decisions were clearly noted in the visuals, and the same rules were applied across all graphs to keep the results consistent and easy to understand.

Items That Still Need Clarification

After the in-depth analysis, there are certain areas that still require additional clarification. Some of the rural counties did not fully report their childcare cost, which could change the outcome of the current stats. The NDCP dataset lacks details on household composition, family income, and childcare subsidies, which limits understanding of the true financial impact on individual families. Incorporating such variables in future analyses would provide more information about the affordability and economic stress.

Direction of the Story

The main goal of this project is to present a data-driven story about affordability and inequality in the U.S. The message highlights that childcare is not just an issue for the families using this service but a national economic issue that impacts workforce participation and community well-being. Similarly, understanding childcare pricing through data can help leaders create a more targeted and fair action plan.

Target Audience

The target audience includes policymakers, local officials, and community organizations that focus on childcare support and family welfare. Secondary audiences include parents, educators, and journalists, who can use the findings to inform decision-making and raise public awareness. Presenting data in multiple formats ensures that both professional and general audiences can access and interpret the findings effectively.

Mediums Included and Rationale

There will be three different mediums created to reach these audiences. A PowerPoint presentation summarizes the key findings with data-driven graphs and captions for policymakers and professionals. A community poster provides an accessible, visually engaging overview for public spaces. Finally, a blog post offered a more detailed, narrative-driven analysis for broader audiences, embedding charts and explanations to enhance the user's understanding. Using multiple mediums allowed the project to explain the dataset in different ways for any audience.

Design Decisions

There were a few different design choices made for this project to keep the visuals consistent and easy to understand. Warm colors such as peach and orange were used across all mediums to create a friendly and positive tone. Each design was kept clean and direct, with limited text, so the audience wouldn't feel so overwhelmed. The PowerPoint had clean, consistent slides with stock images and graphs made with the data, to keep the audience engaged. The poster used an infographic layout, and the blog used embedded visuals for digital readers.

Data Changes and Transformation

During cleaning, column names were standardized, symbols were removed from numeric fields, and missing values were imputed or excluded where necessary. Cost data were converted into percentages of median household income to measure affordability consistently across states and age groups. Extreme outliers were filtered to prevent distortion of the visualizations, and data were aggregated at the state level to highlight overall patterns.

Legal Guidelines

The NDCP dataset is publicly available and sourced from the U.S. Department of Labor, Women's Bureau. No personally identifiable information was included, and all transformations maintained compliance with data privacy standards. Open source data policies were followed, and proper attribution was given to the original dataset.

Data Verification

The NDCP dataset came from a reliable government source and was checked for accuracy against other reports from the U.S. Department of Labor. No personal family information was included, and the analysis was done fairly to show accurate trends at the state level.

Ethical Considerations

The childcare dataset was cleaned carefully to maintain accuracy and privacy. The childcare data was grouped by state, as there were no personal identifiers used during this process. This allowed the families to stay anonymous, but to look at which states need immediate attention.

The dataset was shared publicly and open-sourced. The project shows where the dataset comes from and the changes that have been made for transparency. Any risks from transformations, such as grouping by state or highlighting affordability rankings, were reduced by adding notes and captions to clarify the meaning of each visualization.

The dataset was ethically sourced and verified through reputable sources, including the U.S. Department of Labor and open government repositories. Explanations were added to help the audience understand what the data represented and to prevent misinterpretation.

Lesson Learned

This project showed that combining data analysis with clear design is key to explaining complex social issues. Even small choices, like chart colors, labels, and layout, can affect how people understand the results. It reinforced that telling a story with data requires both accuracy and clarity.

Future Recommendations

If this project were done again, looking at neighborhood level data could show which communities are most affected. An interactive dashboard would let users explore the data themselves. Adding stories or quotes from families would make the findings more relatable and meaningful.

Reflection

This project showed how valuable it is to use different forms of media to share findings with an audience. It also revealed how time-consuming it can be to design each one accurately,

cleanly, and consistently across all platforms. Every design choice, such as color, layout, or labeling, can change how people interpret the results. Paying close attention to the small details can keep the message clear and impactful.

If I were to do this project again, I would look at specific neighborhoods for both datasets. This will allow us to dive deeper into where there are troubled neighborhoods that are highly impacted by this problem and need immediate help. I would also create an interactive dashboard that lets the user explore the data themselves. Including real-life quotes or short stories from families to make the topic even more personal and relatable.

The part that I enjoyed the most was coding out the graphs and then making the poster. I am not the most visual and creative person, but it is nice to have more practice making these visualizations. I am seeing in the workplace, especially with all the new AI tools, they want us to be more visually creative with our findings. Being able to take it that extra step, without handing it off to someone else to do.

Conclusion

Overall, this project used data analysis, design, and storytelling to express a complex issue and show the intended audience that they need to take immediate action. Childcare affordability affects families, communities, and the economy. Using NDCP data and visualizations, this project highlights differences by age and state, showing where policy changes are needed most. Reinforcing the importance of clear communication and thoughtful design when presenting information that can influence real-world decisions.

Resources

U.S. Department of Labor, Women's Bureau. (2024). *National Database of Childcare Prices: 2008-2022*. [Data set] <https://www.dol.gov/agencies/wb/topics/featured-childcare>